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Introduction to NLP

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Learning outcomes

- By the end of this course the learning outcomes are:
    - Define NLP in the broad sense
    - Understand different NLP applications
    - Hands-on experience with text classification and sentiment analysis

OVERVIEW

1. Definition
2. NLP Applications
3. Guided Exercise
    - Problem statement
    - Data Inspection
    - Data Cleaning
    - Data Preprocessing
    - Bag of Words Transformation
    - Classification with Logistic Regression

Definition

Natural language processing (NLP) is the ability of a computer program to understand human language as it is spoken and written

Challenges of Text Data

- High dimensionality, sparsity, ambiguity.
- Different languages, syntax, and semantics.
- Context and meaning variation.
- Handling informal language, slang, and domain-specific terminology.

NLP Applications

- Text Classification
- Sentiment Analysis
- Chatbots & Virtual Assistants
- Named Entity Recognition (NER)

NLP Applications

- Speech Recognition
- Machine Translation
- Text Summarization
- Auto-Correct

Text Preprocessing

- Improves model performance by standardizing input data and reducing noise.
- Makes text representation more consistent and suitable for analysis.

Basic Text Processing Steps

- **Tokenization** - splitting text into words/subwords, sentence-level tokenization.
- **Stopword Removal** - removing common but unimportant words.
- **Lowercasing** - normalizing text to improve consistency.
- **Stemming vs. Lemmatization** - reducing words to their base forms.
- **Removing Punctuation & Special Characters** - improving data quality and reducing noise.

Guided Exercise

Problem Statement

- Goal: Classify users' reviews as **good** or **bad** reviews'
- Dataset used: Amazon Fine Food Reviews Dataset

Data Inspection

The process of loading and viewing the data for verification purpose

Data Inspection - Exercise

Use pandas to:

- Load the **Reviews.csv** file into a pandas dataframe
- Drop the rows containing null values
- Print the shape of the dataset
- Print the first 2 rows of the dataset

Data Cleaning

Data cleaning is the process of fixing or removing incorrect, corrupted, incorrectly formatted, duplicate, or incomplete data within a dataset

Data Cleaning - Exercise 1

Use pandas and matplotlib to:

- Remove the duplicate rows, which share the same UserId, ProfileName, Time, and Text

- Create a histogram showing the review distribution over scores from 1 to 5

Data Cleaning - Exercise 2

Use pandas to:

- Create a label column in the dataframe where:
  - label = 0 if score <= 3
  - label = 1 if score > 3

Text Preprocessing

Text preprocessing is the first step of NLP projects and its purpose is to prepare the data for the NLP model

Text Preprocessing

Common text preprocessing techniques include:

- Decontraction
- Punctuation/URL Removal
- Stop words removal
- Lowercasing
- Tokenization
- Stemming
- Lemmatization

Text Preprocessing - Exercise 1

Decontraction is the process of replacing contracted words with their longer form. *ex: "aren't" → "are not"*

Write a function called **decontract_words** that takes a text as input, splits it and decontracts the contracted words using the provided contraction dict. The function should re-assemble the text and return it

Text Preprocessing

- Punctuation and URL removal is done through regular expressions **(Regex)**

- Regular Expression, is a sequence of characters that forms a search pattern. Regex can be used to check if a string contains the specified search pattern.

Text Preprocessing - Exercise 2

Create a function that uses regular expressions and python re library to:

- Remove all URLs
- Remove html style link (<a href=...)
- Remove all &
- Remove special characters
- Remove all <br/ >

Text Preprocessing - Exercise 3

Create a function that uses **NLTK english stopwords** to remove stopwords from text and returns the text

Text Preprocessing

- Lowercasing means converting all chars to lowercase

- Tokenization is the process of breaking down sentences into words

Text Preprocessing - Exercise 4

Create a function that performs all preprocessing operations by:

- Calling previously created preprocessing functions
- Apply lowercasing to text
- Tokenize text using **nltk WordPunctTokenizer**
- Apply this function to the text column dataframe and create a new column to store them called text_cleaned

Text Preprocessing

- Stemming and Lemmatization are techniques that diminish words to their root/base form

Original Word	After Stemming	After Lemmatization
goose	goos	goose
geese	gees	goose

Text Preprocessing - Exercise 5

- Apply lemmatization to text_cleaned column using **nltk WordNetLemmatizer**

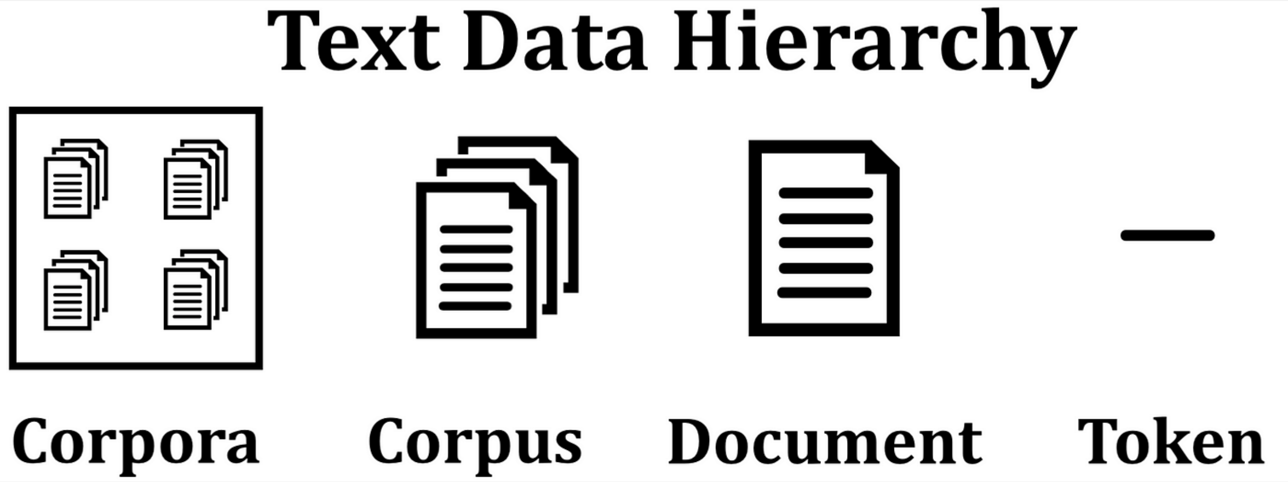
- Store the lemmatized data in a new column in the dataframe called text_lemmatized

Text Vectorization

- Text Vectorization is the process of converting text into numerical representation.

- Popular vectorization techniques:
    - Bag of Words (BoW)
    - Term frequency , Inverse document frequency (TF-IDF)
    - Word2Vec/Doc2Vec
    - GloVe

Text Vectorization - Terminologies



- Document: each text sentence
- Corpus: collection of text documents

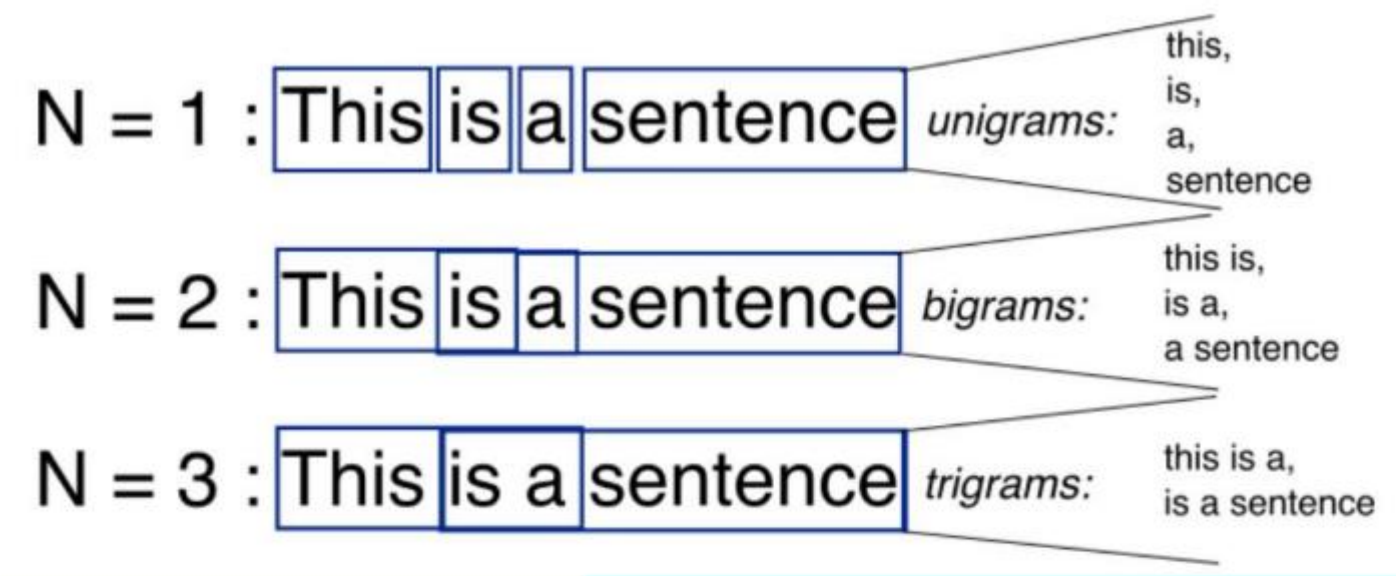
Text Vectorization - BoW

The intuition behind BoW is that two sentences are said to be similar if they contain a similar set of words.

	about	bird	heard	is	the	word	you
About the bird, the bird, bird bird bird	1	5	0	0	2	0	0
You heard about the bird	1	1	1	0	1	0	1
The bird is the word	0	1	0	1	2	1	0

Text Vectorization - N-Grams

- N-Grams are a set of co-occurring words



N-grams can be used when we want to preserve sequence information in the document, like what word is likely to follow the given one.

Text Vectorization - Exercise

- Use scikit's **train_test_split** to split the dataset 80% for training and 20% for testing

- Use scikit's **CountVectorizer** to create a BoW representation of the data:
  - *Hint 1: Use lambda doc: doc as tokenizer*
  - *Hint 2: **CountVectorizer** should allow single words, bi-grams and tri-grams*

Text Classification - Exercise

- Use scikit's **Logistic Regression** to train a classification model and test this model with the testing data you partitioned and print the model's score

Text Vectorization - TF-IDF

Term Frequency – Inverse Document Frequency

Determines the importance of a term in a text and gives it a weight in proportion with that importance.

$$\text{TF}(\text{word}, \text{text}) = \frac{\text{number of times the word occurs in the text}}{\text{number of words in the text}}$$

$$\text{IDF}(\text{word}) = \mathbf{log}\left[\frac{\text{number of texts}}{\text{number of texts where the word occurs}}\right]$$

$$\text{TF-IDF}(\text{word}, \text{text}) = \text{TF}(\text{word}, \text{text}) \times \text{IDF}(\text{word})$$

Text Vectorization - TF-IDF

- Review 1: This movie is very scary and long
- Review 2: This movie is not scary and is slow
- Review 3: This movie is spooky and good

Term	Review 1	Review 2	Review 3	IDF	TF-IDF (Review 1)	TF-IDF (Review 2)	TF-IDF (Review 3)
This	1	1	1	0.00	0.000	0.000	0.000
movie	1	1	1	0.00	0.000	0.000	0.000
is	1	2	1	0.00	0.000	0.000	0.000
very	1	0	0	0.48	0.068	0.000	0.000
scary	1	1	0	0.18	0.025	0.022	0.000
and	1	1	1	0.00	0.000	0.000	0.000
long	1	0	0	0.48	0.068	0.000	0.000
not	0	1	0	0.48	0.000	0.060	0.000
slow	0	1	0	0.48	0.000	0.060	0.000
spooky	0	0	1	0.48	0.000	0.000	0.080
good	0	0	1	0.48	0.000	0.000	0.080

Text Vectorization - TF-IDF

Use scikit's TFIDF Transformer on top of the CountVectorizer to create TF-IDF vectors for your existing BoW reviews vectors

Text Vectorization - Limitations

BoW

Creates a set of vectors containing the count of word occurrences in the document.

>> easy to interpret

TF-IDF

Contains information on the more important words and the less important ones as well.

>> performs better in machine learning models

→ Remained void: understanding the context of words.

No detecting the similarity between the words ‘spooky’ and ‘scary’, cannot carry semantic meaning

→ Ignores word order and negation cases

“Queen of England” will not be considered as a single unit

“not pay the bill” vs “pay the bill” makes no difference

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THANK YOU

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